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Improving Travel Mode Choice Predictions for Minority Modes with a Conditional Variational Autoencoder to Address Class Imbalance

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Abstract

Class imbalance in travel mode choice data represents a significant challenge to transportation modeling, often leading to biased parameter estimates and high prediction errors for less frequently chosen alternatives. This study presents a comparative analysis of the Multinomial Logit (MNL) model's predictive performance when class imbalance is addressed using Random Oversampling (ROS), Synthetic Minority Oversampling Technique (SMOTE), class-weighting, and the Conditional Variational Autoencoder (CVAE). To the best of our knowledge, this is the first application of CVAE to address class imbalance in travel mode choice prediction. Using three well-known datasets, we estimate the MNL model on both imbalanced and balanced data. Our results show that CVAE improves predictive performance only under severe class imbalance and only in one of three datasets, while performing worse than the unbalanced baseline in another. More critically, CVAE introduces sign reversals in key behavioral parameters in two of the three datasets. These reversals are not inherited from the original data but are specific to the CVAE augmentation method, and they render the resulting model economically implausible for policy analysis. We trace these reversals to a concrete mechanism: the distortion of attribute gaps between chosen and unchosen alternatives in the synthetic data. In contrast, ROS and class-weighting preserve behavioral plausibility while providing competitive predictive performance. These findings highlight a fundamental trade-off between predictive accuracy and behavioral interpretability under generative data augmentation. CVAE should be restricted to purely predictive applications, and its use must be accompanied by verification of coefficient coherence before any policy interpretation is attempted.

Keywords

Conditional Variational Autoencoder, Travel Mode Choice, Multinomial Logit, Class Imbalance, Behavioral Interpretability

Introduction

Predicting travel mode choice accurately is one of the primary goals of transportation planners Rezaei et al. (2021). According to Ortúzar and Willumsen (2011), mode choice analysis helps planners understand the relative importance of individual and contextual factors shaping travel preferences. The Multinomial Logit Model (MNL) is the most common model for understanding mode choice. However, logit models usually output biased estimates and have large prediction errors for less commonly used modes Rezaei et al. (2021). Traditionally, the performance of logit models is evaluated based on behavioral analysis, privileging improvements in goodness-of-fit, while less attention has been paid to improving predictive ability Mullainathan and Spiess (2017). According to Rezaei et al. (2021), although behavioral analysis of mode choice models is essential for improving parameter estimation, enhancing predictive capability should not be ignored. As Chen and Cheng (2023) indicate, class imbalance severely compromises model estimation and predictive performance, as the model tends to focus on the majority class while ignoring minority classes. This may affect the formulation of policies underlying the estimation of this class of models.

Datasets used for estimating mode choice models typically exhibit substantial differences in the number of observations across alternatives (Burbidge and Goulias 2009). Such imbalance arises when one or more classes contain considerably fewer observations than others (Cicek et al. 2023). Despite its relevance, class imbalance remains comparatively underexplored in the discrete choice modelling literature. In the context of choice modeling, logit models often produce biased estimates and large prediction errors for less frequently used modes (Rezaei et al. 2021). The minority category, including the incipient use of bicycles in certain cities or public transport in areas with limited

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coverage or relatively new services, can present significant challenges in predicting travel demand (e.g., Whalen et al. 2013; Hagenauer and Helbich 2017; Lee et al. 2018). These problems can permeate transportation planning and lead to systematic errors in infrastructure investment and policy design.

The policy consequences of minority mode underrepresentation are concrete. As noted by Wang et al. (2023), bicycle use surged in the post-COVID-19 era due to its suitability for socially distanced travel and environmental benefits. However, the persistent underrepresentation of cycling in traditional travel surveys often leads to demand underestimation, limiting investment in micromobility infrastructure and emissions reduction strategies. More broadly, when models systematically underestimate the demand for less common travel modes, this may lead to underinvestment in infrastructure, misallocation of resources, and the neglect of specific population needs, reinforcing unsustainable mobility patterns and hindering efforts to promote equitable and inclusive transportation systems.

In the machine learning (ML) literature, several works have studied the predictive performance of ML algorithms and discrete choice models Hagenauer and Helbich (2017); Wang and Ross (2018); Zhao et al. (2019); Salas et al. (2022). However, both methodologies suffer from biases and accuracy problems when the dataset is highly imbalanced van Cranenburgh et al. (2022). Several methods based on sampling techniques address the imbalance problem — for example, Random Oversampling (ROS) and the Synthetic Minority Oversampling Technique (SMOTE) Chawla et al. (2002). However, these methods are limited to resampling existing observations and cannot generate genuinely new data that captures the underlying distribution of the minority class. According to Islam et al. (2021), the main limitation of oversampling methods includes creating unwanted samples around the decision boundary of the majority and minority classes, disrupting the natural border between classes.

From ML, generative models offer an alternative for overcoming the limitations of small or imbalanced datasets. One of the most popular generative models is the Variational Autoencoder (VAE) Kingma and Welling (2013); Rezende et al. (2014). The VAE is an unsupervised learning model that combines Bayesian inference and neural networks to obtain a nonlinear low-dimensional latent space with suitable properties for the generative process Zemouri et al. (2019). In data generation with VAE, the decision boundary can be precisely selected, reducing overfitting inherent to synthetic oversampling methods Islam et al. (2021). However, generating class-specific samples is not possible with a standard VAE Yang et al. (2019). To address this limitation, Kingma et al. (2014) proposed the Conditional Variational Autoencoder (CVAE), which extends the VAE into a supervised generative model. In the context of data augmentation, Lopez-Martin et al. (2017), Gong et al. (2020), Mu and Chen (2022) and Fan et al. (2022) show that CVAE enables improved prediction accuracy of classification models.

This study proposes and evaluates the use of CVAE as a data augmentation strategy to improve the predictive performance of the MNL model in the travel mode choice context with imbalanced data, and compares it against ROS, SMOTE, and class-weighting. Our analysis reveals that while CVAE can improve predictive metrics under severe class imbalance, it introduces sign reversals in key behavioral parameters (travel time, waiting time, and travel cost) that compromise the economic interpretability of the estimated model. We trace these reversals to a distortion of the structural relationship between attributes and choices in the synthetic data, and show that this distortion does not arise under any other method evaluated. To the best of our knowledge, this is the first application of CVAE to address class imbalance specifically in travel mode choice prediction.

The rest of this article is structured as follows. Literature Review presents a brief review of the literature. Methods describes the methods, followed by the description and results of the empirical application in Application. Finally, Discussions and Conclusions close the paper with concluding remarks, limitations, and directions for future research.

Related Works

The evaluation of mode choice models generally relies on two complementary dimensions. The first concerns behavioral interpretability, which refers to the model's capacity to adequately represent and explain observed travel behavior. The second relates to predictive performance, which refers to the model's ability to generalize beyond the estimation sample. As noted by Rezaei et al. (2021), the literature has traditionally prioritized improvements in model fit, placing comparatively less emphasis on out-of-sample predictive accuracy. Nevertheless, several studies argue that explanatory power and predictive capability should be considered equally relevant in travel behavior research (e.g., Hagenauer and Helbich 2017; Wang and Ross 2018; Zhao et al. 2020; Salas et al. 2022; Rezaei et al. 2021). In this context, van Cranenburgh et al. (2022) suggest that discrete choice modelling could benefit from adopting evaluation practices commonly used in machine learning, particularly those focused on rigorous assessment of statistical predictive performance. Crucially, these two dimensions are not always aligned: a method that improves predictive performance on a holdout sample does not necessarily produce a better discrete choice model in behavioral terms, particularly when the holdout shares the same imbalance and correlation structure as the training data.

In general, the number of observations for various modes in travel mode choice datasets is significantly different. Mode choice models tend to underestimate less commonly used modes, resulting in high prediction error Wang and Ross (2018). Moreover, according to Elamrani Abou Elassad et al. (2024), when handling imbalanced datasets, accuracy may suffer due to bias toward the majority class. This problem is particularly acute because a model can achieve high overall accuracy by systematically ignoring minority classes — a failure mode that standard accuracy metrics do not capture. For instance, in our own Swissmetro dataset, the unbalanced MNL achieves an accuracy of 0.576 but a recall of 0.000 for the minority class: the model never predicts the train alternative despite it representing 13% of the sample. Accurate prediction across all travel modes is nonetheless equally essential for informed future planning (Rezaei et al.

2021), since underestimating minority modes may introduce systematic errors in infrastructure investment and policy design (Kumar et al. 2021).

In several recent papers related to mode choice, the predictive ability of logit models has been analyzed and compared with machine learning algorithms. The impact of class imbalance on predictive ability is significant, and prior work reports a wide range of accuracy values across different datasets and class distributions. Interpreting these figures requires distributional context: a model that achieves high accuracy on a dataset where one alternative accounts for 80% of observations may simply be predicting the majority class for every observation. Wang and Ross (2018) suggest avoiding extremely imbalanced datasets by collecting more samples for minority travel modes. Alternatively, applying resampling techniques may help improve model performance in predicting travel mode choice. Salas et al. (2022) provides a systematic comparison using well-known datasets in the travel mode choice literature, showing that accuracy is strongly influenced by the degree of class imbalance.

In other fields, SMOTE Chawla et al. (2002) and ROS are the most popular techniques for balancing and augmenting datasets in classification problems. However, according to Islam et al. (2021), the main limitation of oversampling methods includes creating unwanted samples around the majority and minority class decision boundary, altering the natural border between classes. Several authors argue that generative ML models can overcome these limitations Islam et al. (2021); Chalé and Bastian (2022). Moreover, Chalé and Bastian (2022) indicate that generative models go beyond sampling techniques to model the entire joint distribution of the data, enabling the synthesis of realistic examples from complex distributions.

The use of generative models, such as the VAE proposed by Kingma and Welling (2013); Rezende et al. (2014), has been shown to improve model performance Perez and Wang (2017). VAEs are probabilistic frameworks that employ variational Bayesian inference to approximate the latent generative distribution (Yao and Bekhor 2022). Previous researchers have reported that VAE achieves great success in improving classification accuracy (for example, Lopez-Martin et al. 2017; Zhao et al. 2019; Islam et al. 2021). Also, Moreno-Barea et al. (2020) and Garay-Maestre et al. (2018) showed that models built after data augmentation with VAE achieve better performance on small datasets. In the transport field, Yao and Bekhor (2022) use the VAE approach for route choice set generation, and Liu et al. (2024) propose data- and knowledge-driven strategies to enrich discrete choice model formulations. However, the standard VAE does not allow control over the class of generated samples. Conditional Variational Autoencoders (CVAEs) (Kingma et al. 2014) address this by incorporating class labels into the generative process (Ramchandran et al. 2022). The benefits of CVAE for prediction have been demonstrated in vehicle trajectory prediction, intrusion detection, electricity theft detection, and pathology detection in medical images Feng et al. (2019); Uzunova et al. (2019). In the context of data augmentation, Lopez-Martin et al. (2017), Gong et al. (2020), Mu and Chen (2022) and Fan et al. (2022) show that CVAE enables improved prediction accuracy of classification models. De Donno et al. (2023) apply CVAEs to learn representations for data integration and annotation tasks.

While this study applies CVAE primarily as a data augmentation technique to address class imbalance in travel mode choice prediction, recent research has explored the broader potential of autoencoder-based architectures in discrete choice modeling. Yao and Bekhor (2020, 2022) and Liu et al. (2024) propose variational autoencoders not only to enhance predictive accuracy, but also to support choice set generation and model the implicit perception of alternatives. More recently, Wang et al. (2023) applied a combination of oversampling techniques and machine learning classifiers to address class imbalance in travel mode choice, providing a useful reference point for the present study. Additionally, Li and Kockelman (2022) provide a systematic comparison of machine learning and econometric approaches for transport data, including an analysis of how predicted probabilities vary across the sample — an approach we adopt in Section to diagnose the behavioral consequences of sign reversals. These contributions highlight that generative models can be leveraged not only for improving classification performance but also for structurally enriching discrete choice model formulations.

To the best of our knowledge, there has been limited attention in the literature regarding class imbalance in travel mode choice prediction. The works of Rezaei et al. (2021) and Chen and Cheng (2023) stand out: Rezaei et al. (2021) assessed the impact of resampling techniques on logit model performance, while Chen and Cheng (2023) explored hybrid statistical and machine learning models alongside multiple resampling methods to boost prediction accuracy.

In this paper, we evaluate SMOTE, ROS, class-weighting, and CVAE for travel mode choice data balancing to improve the predictive performance of the MNL model, and examine the behavioral consequences of each method. To the best of our knowledge, this is the first study to evaluate the impact of CVAE specifically on class imbalance correction in travel mode choice prediction, and to document the trade-off between predictive performance and behavioral plausibility under generative data augmentation.

Methods

In this section, we define the main methods used in this work. First, we present the MNL model, which is used to study the effects of balancing the dataset using the different methods on the predictive performance. Second, we present a brief description of traditional oversampling methods. Then, to better understand Conditional Variational Autoencoder (CVAE), we first describe the Variational Autoencoder (VAE). Finally, we present the semi-supervised ML model called CVAE, an extension of VAE that adds response (choice) information in the model training process.

Multinomial Logit

The Multinomial Logit (MNL) model (McFadden 1973) is derived from the Random Utility Maximization (RUM) framework, which assumes that the indirect utility U_{nj} for individual n and alternative j is observed by the decision-maker but not by the analyst (Castro et al. 2013). Utility is

decomposed into a systematic component, $V_{nj}(\boldsymbol{\beta})$, defined as a function of observed attributes and parameters, and a random error term, ϵ_{nj}. Thus, the indirect utility is expressed as:

$$U_{nj} = V_{nj}(\boldsymbol{\beta}) + \epsilon_{nj} \quad . \tag{1}$$

The MNL model assumes that the error term ϵ_{nj} is independently and identically distributed (i.i.d.) according to a Gumbel distribution, which yields a closed-form expression for the choice probabilities. Accordingly, the probability that individual n selects alternative j is given by:

$$P_{nj} = \frac{e^{V_{nj}}}{\sum_{i=1}^{J} e^{V_{ni}}}. \tag{2}$$

Under the assumption that the error terms are independently and identically distributed, and given the parameter vector $\boldsymbol{\beta}$, the joint density of all observed individual choices in the MNL framework can be expressed through the following likelihood function:

$$\begin{aligned} \mathbf{L}(\boldsymbol{\beta}) &= \prod_{n=1}^{N} \prod_{j=1}^{J} \left[\frac{e^{V_{nj}(\boldsymbol{\beta})}}{\sum_{j=1}^{J} e^{V_{nj}(\boldsymbol{\beta})}} \right]^{y_{nj}} \\ &= \prod_{n=1}^{N} \prod_{i=1}^{J} (P_{nj}^{y_{nj}}), \end{aligned} \tag{3}$$

where y_{nj} is equal to one if the individual n chooses alternative j and 0 otherwise.

The parameter vector $\widehat{\boldsymbol{\beta}} = \arg\max_{\boldsymbol{\beta}} \mathbf{LL}(\beta)$ is obtained via maximum likelihood estimation. In this study, the MNL model is estimated using the Apollo R package (Hess and Palma 2019), employing the Broyden–Fletcher–Goldfarb–Shanno (BFGS) optimization algorithm.

Variational Autoencoder

Variational autoencoder (VAE), initially proposed by Kingma and Welling (2013) is a type of generative model, which aims to learn the distribution of the input data x Mancisidor et al. (2021). This means that the VAE can sample from a distribution that it is similar to the one that have generated x Mancisidor et al. (2021). In general, the VAE is composed of the same three elements as an autoencoder: 1) encoder, 2) latent space, and 3) decoder. According to Islam et al. (2021), the encoder mapping the input x to a set of parameters that completely define an intermediate probability distribution: $q_\phi(z|x)$. Then, a latent space z is generated from samples from this intermediate distribution. Finally, z form the input to the decoder. The decoder works similarly to the encoder, mapping from z to a new set of parameters defining a new probability distribution: $p_\theta(\hat{x}|z)$, from which we take samples again. These samples make up the output of the VAE: $\hat{x}$.

Using a variational approach Kingma and Welling (2013) VAEs learn the conditional probability distributions $q_\phi(z|x)$ and $p_\theta(\hat{x}|z)$. This approach moves the learning process to a minimization process, which can be formulated in terms of the stochastic gradient descent algorithm Lopez-Martin et al. (2017). The subscripts θ and ϕ represent the set of parameters of the neural networks that compose the encoder and decoder, respectively. These parameters are tuned as part of the VAE training process and are considered constant later on Lopez-Martin et al. (2017). Another particularity of VAE is that instead of directly generating a latent vector and maximizing the marginal log-likelihood, a vector of means μ and a vector of standards deviations σ are generated, and they are combined to form the latent vector. However, according to Moreno-Barea et al. (2020) the direct combination of these parameters in the latent vector z implies that this is a continuous random variable, which would not allow the network to learn the distribution that comes from the encoder. In order to solve this problem and express the random variable z as a deterministic variable, Kingma and Welling (2013) proposed a simple method known as "parameterization trick". In this method, z is expressed as a deterministic function of the encoder output parameters $(\mu; \sigma)$ and a stochastic term ϵ sampled from a standard normal distribution. The parameterization trick is shown in Eq. (4)

$$z = \mu + \sigma\epsilon, \quad \epsilon \sim N(0, 1). \tag{4}$$

In the variational approach, the main idea is to maximize the probability of obtaining the desired data as output by maximizing a quantity known as the Evidence Lower Bound (ELBO) Lopez-Martin et al. (2017). In ML fields, the ELBO represents the loss function of the VAE model. The loss function $ELBO$ (Eq. (5)) consist the two terms; the first terms penalizes the reconstruction error between the input x and the output $\hat{x}$ and the second term penalizes the error between the prior distribution of latent space $p_\theta(z)$ and the learned distribution $q_\phi(z|x)$. Figure 1 shows a diagram representing the schematic representation of basic VAE structure.

$$ELBO = \log p_\theta(x|z) - D_{KL}(q_\phi(z|x)||p_\theta(z)), \tag{5}$$

where the term $D_{KL}(q_\phi(z|x)||p_\theta(z))$ is the Kullback–Leibler divergence between the distribution of the encoder $q_\phi(z|x)$ and the prior distribution of latent space $p_\theta(z)$.

Conditional Variational Autoencoder

The conditional variational autoencoder (CVAE) Kingma et al. (2014) is a variant of Variational Autoencoder, this model focuses on conditional data generation using semi-supervised learning. In general, the CVAE is composed of the same three elements as an autoencoder: 1) encoder, 2) latent space, and 3) decoder. In the CVAE, the encoder approximates the generative distribution of the latent variables using the data (x) and the labels (y). In contrast, the decoder approximates the generative distribution of the data when given the latent variables z and y. Having the labels as extra input leads to the probability distribution of the latent space being conditioned by x and y, while the probability distribution of the decoder is conditioned by z and by y. In Figure 2 shows a diagram representing the schematic representation of basic CVAE structure. The label/response y is added in both the encoder and the decoder network. Thus, CVAE can generate conditional outputs accordingly, which

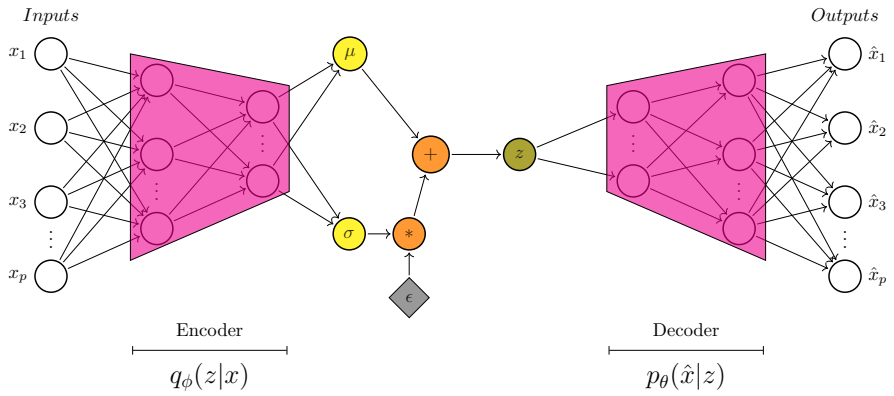


Figure 1. Schematic representation of Variational Autoencoders architecture. The stochastic component ϵ in the gray diamond is sampled from normal standard distribution.

offers a more interpretable approach to produce a multi-modal prediction than other methods Feng et al. (2019).

CVAEs have been proposed in the literature with different architectural choices depending on the learning objective. In our study, we adopt the formulation where both the encoder and decoder are conditioned on the class label,

$$q_\phi(z|x,y) \quad \text{and} \quad p_\theta(x|z,y), \tag{6}$$

following the formulation proposed by Sohn et al. (2015). This setup is particularly appropriate for our goal, which is to generate synthetic alternatives for specific classes in discrete choice datasets—an application that requires fine control over the output class of the generated data. Other variants of CVAEs have been proposed. For example, Ivanov et al. (2018) extended this idea with more flexible conditioning mechanisms, including hierarchical or implicit conditioning strategies. While such alternatives are advantageous in tasks like semi-supervised classification or disentangled representation learning, our objective differs: we aim to augment class-specific samples in a controlled and interpretable way to mitigate class imbalance in discrete choice modeling. Conditioning both the encoder and decoder on the class label ensures that the latent representations are disentangled by class and that the generated data remains consistent with the class-specific structure of the original dataset.

In Figure 2 we present the structure used in this work, specifically, both the encoder and decoder architectures can be composed of one or multiple hidden layers.

As mentioned, CVAE adds the y response in the encoder and decoder part of VAE. Thus the ELBO function present in Eq. (5) takes the following form

$$ELBO = \log p_\theta(x|z,y) - D_{KL}(q_\phi(z|x,y)||p_\theta(z)), \tag{7}$$

If the CVAE is adequately trained, one can use the decoder part of the network to generate any number of samples of a particular class, realizing a sampling of z from a standard normal distribution, concatenate with respective value of y, and pass it through the decoder Mishra et al. (2018). In this work, the conditional y are the travel mode alternatives.

However, when employing models with variational approaches, it is common to observe overlapping latent representations between classes, which may hinder the generation of high-quality synthetic samples. As noted by Islam et al. (2021), relying solely on visual inspection of the latent space can lead to imprecise or biased sampling regions, particularly when the boundaries between classes are not well defined.

Latent Space Sampling and Synthetic Data Generation

In CVAE, the posterior distribution of the latent space is already conditioned on the class label, so the latent representations of different classes are not required to be well-separated. In practice they are not, as we show in Section . This is a key property of the CVAE architecture that distinguishes it from the standard VAE.

Nevertheless, sampling uniformly from the full latent space can produce low-quality synthetic observations, particularly in the tails of the learned distribution where the generative model has seen few training examples and data density is low. To avoid this, we restrict sampling to high-density regions of the latent space by constructing 80% confidence ellipses for each travel mode. These ellipses serve as a practical sampling mechanism, not as a solution to class separation: their purpose is to concentrate synthetic observations in the region where the CVAE has reliable generative capacity, rather than to impose boundaries between classes.

To construct these ellipses, we first encode the observed alternatives into a two-dimensional latent space using the trained encoder of the CVAE. Each encoded point corresponds to a latent representation conditioned on its class label. For each class (i.e., travel mode), we calculate the mean and covariance of the latent representations and construct confidence ellipses using the method proposed by Friendly et al. (2013). Sampling within each ellipse is performed stochastically: candidate points are drawn from the multivariate normal distribution of each class and accepted if their Mahalanobis distance falls within the ellipse boundary, ensuring that the density of synthetic observations follows that of the real data rather than being uniform across the ellipse.

We use two latent dimensions (Z_1 and Z_2) for ease of visualization and generation. The sensitivity of the results to

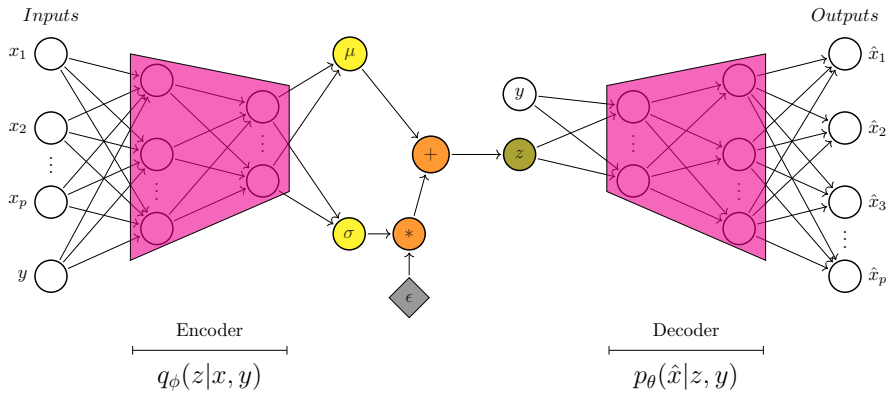


Figure 2. Schematic representation of Conditional Variational Autoencoders architecture. The stochastic component ϵ in the gray diamond is sampled from normal standard distribution.

the choice of confidence level (70%, 80%, 90%, and 95%) is examined in Section .

Oversampling Techniques

Oversampling techniques aim to address class imbalance by increasing the representation of the minority class. Two common approaches are Random Oversampling (ROS) and Synthetic Minority Oversampling Technique (SMOTE). While ROS simply duplicates existing instances of the minority class Abou Elassad et al. (2020), SMOTE generates new, synthetic samples by interpolating between existing ones Chawla et al. (2002).

Specifically, SMOTE begins by identifying the k-nearest neighbors of each instance in the minority class. Based on the desired amount of oversampling, it randomly selects one of these neighbors and generates a new synthetic observation along the line segment connecting the two instances. This is achieved by computing the difference between the feature vectors of the selected pair and multiplying it by a random scalar λ drawn from a uniform distribution in the interval $[0, 1]$. The new synthetic instance is then calculated as follows:

$$x_{\text{new}} = x_i + \lambda \cdot (x_{\text{nn}} - x_i)$$

where x_i is an instance from the minority class, x_{nn} is one of its k-nearest neighbors, and λ is the random scalar. This interpolation strategy ensures that the new sample remains within the local neighborhood of the minority class distribution, enriching the feature space without merely duplicating existing data. Finally, the generated instance is labeled as part of the minority class and added to the training dataset.

Class-weighting is an alternative approach that does not generate synthetic observations. Instead, it assigns each training observation a weight inversely proportional to its class frequency, penalizing misclassification of minority modes more heavily during MNL estimation. This method is included as a baseline that addresses class imbalance without modifying the training data distribution.

Model Evaluation Metrics

We evaluate the prediction capability to assess the balancing effect and generate new samples for a travel mode choice datasets using CVAE in the logit models domain, specifically considering the MNL model. According to Rezaei et al. (2021), the predictive performance of logit models can be evaluated at two levels: individual and aggregate. The former assesses the accuracy of predictions for each travel mode at the individual level, whereas the latter examines the model's ability to reproduce overall mode shares. Notably, logit models are primarily designed to replicate aggregate choice distributions, particularly market shares (Ortúzar and Willumsen 2011; Zhao et al. 2020; Rezaei et al. 2021). In this paper, we focus on individual-level performance. The predicted performance is tested once the deterministic labeling is obtained; i.e., the prediction that is made using the empirical model Japkowicz and Shah (2011). Generally, the deterministic label of mode (y_i) for each individual is based on the largest mode probability Ortúzar and Willumsen (2011); Train (2009), i.e,

$$y_i = \underset{k}{\text{argmax}}\{P_{i1}, \ldots, P_{ik}, \ldots, P_{iK}\}. \tag{8}$$

The prediction results can be summarized using a confusion matrix, as illustrated in Table 1 for a problem with K alternatives. Let n_{ij} denote the number of individuals whose true alternative is i but are predicted as j, and let N_j represent the total number of individuals observed in alternative j. Accordingly, the diagonal elements n_{ii} correspond to correctly classified observations for alternative i.

From the confusion matrix, it is possible to obtain indicators at the individual level; specifically, we use accuracy or first preference recovery, sensitivity or precision, and F1-score. These metrics are shown below

$$Accuracy = \frac{\sum_{i=1}^{K} n_{ii}}{N}, \tag{9}$$

$$Sentivity_k = \frac{n_{kk}}{N_k} \quad \forall k \in \{1, \ldots, K\}, \tag{10}$$

$$F1 - Score_k = 2\left[\frac{Sentivity_k \times Recall_k}{Sensitivity_k + Recall_k}\right], \tag{11}$$

where $N = \sum_{i=1}^{K} N_i$ and $Recall_k = \frac{n_{kk}}{\sum_{i=1}^{K} N_{ik}}$.

Table 1. Confusion matrix for a problem with K alternatives.

		Predicted Alternative					
		1	2	3	$\cdots$	K	Total
Observed Alternative	1	n_{11}	n_{12}	n_{13}	$\cdots$	n_{1J}	N_1
	2	n_{21}	n_{22}	n_{23}	$\cdots$	n_{2K}	N_2
	3	n_{31}	n_{32}	n_{33}	$\cdots$	n_{3K}	N_3
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\cdots$	$\vdots$	$\vdots$
	K	n_{K1}	n_{K2}	n_{K3}	$\cdots$	n_{KK}	N_K

In this work, we will pay greater attention to the harmonic mean of precision and recall (F1-score). The F1-score is a metric used in machine learning and statistics to evaluate the predictive performance of models, especially when dealing with imbalanced datasets.

It is worth noting that the argmax-based evaluation framework adopted here, in which each individual is assigned to the alternative with the highest predicted probability, is standard in the machine learning literature on imbalanced classification, but differs from the econometric practice of evaluating models using the average probability of the observed choice. The latter respects the probabilistic nature of the MNL model and is less sensitive to class imbalance. We retain the argmax-based framework for three reasons. First, the central research question of this paper is whether balancing methods improve the model's ability to correctly identify minority travel modes at the individual level. This is inherently a classification question, for which argmax assignment is the natural evaluation criterion. Second, this framework allows direct comparison with prior work on imbalanced classification in travel behavior (e.g., Rezaei et al. 2021; Chen and Cheng 2023). Third, and most importantly, the sensitivity of argmax-based metrics to class imbalance is a feature rather than a limitation in our context: it is precisely this sensitivity that reveals whether augmentation methods succeed in recovering minority class predictions. We complement these metrics with elasticity-based analysis in Section to provide a behavioral assessment that respects the probabilistic structure of the MNL model.

Quantitative Evaluation of Distributional Similarity

To quantitatively assess the similarity between the real and CVAE-generated data distributions, we used the Jensen-Shannon Divergence (JSD). For each variable, we estimated the empirical probability distributions from the original and synthetic datasets using histogram-based binning with consistent bin widths. Let P and Q denote the discrete distributions corresponding to the real and generated data, respectively. The JSD is defined as:

$$\text{JSD}(P||Q) = \frac{1}{2}\text{KL}(P||M) + \frac{1}{2}\text{KL}(Q||M),$$

where $M = \frac{1}{2}(P + Q)$.

JSD is symmetric and bounded between 0 and 1, with lower values indicating higher similarity. All JSD computations were performed independently for each variable, and the results were aggregated across datasets. This approach enables a consistent and interpretable evaluation of how closely the CVAE replicates the statistical properties of the original data.

Application

This section presents the results of this study. To enhance the credibility of our work, we have performed an analysis with three real-world datasets. These datasets have been used in different articles and are open-access.

Datasets

The Swissmetro The Swissmetro dataset is based on a Stated Preference (SP) survey designed to assess the potential demand for a proposed high-speed transportation system in Switzerland, known as Swissmetro (Bierlaire et al. 2001). Due to its relevance and structure, the dataset has been widely used to evaluate discrete choice models (Hess 2005; Hess et al. 2005, 2006, 2010; Sifringer et al. 2020) as well as machine learning approaches (Salas et al. 2022). The choice set comprises three intercity travel alternatives: Car (available only to car owners), Train, and Swissmetro. While Car and Train correspond to existing modes, Swissmetro represents a hypothetical alternative. Additional details on the variables can be found in Bierlaire et al. (2001). The final sample includes 9,036 observations. As reported in Table 2, Swissmetro is the most frequently chosen alternative (58.0%), followed by Car (28.8%) and Train (13.2%).

Las Condes - Centro Corridor The Las Condes–Centro (LC–Centro) dataset was collected in 1983 (see Ortúzar et al. 1983). It comprises 697 individuals from different areas of Santiago, Chile, all of whom are users of a specific transit corridor. The original choice set includes nine travel alternatives, covering both public transport modes (bus, metro, shared taxi, and their combinations) and private modes (car-driver and car-passenger). This dataset has been widely used in the estimation of discrete choice models (e.g., Gaudry et al. 1989; Munizaga and Alvarez-Daziano 2002; Guevara 2017), and more recently, in the application of machine learning approaches (Salas et al. 2022). For illustrative purposes, this study focuses on four alternatives. As shown in Table 2, car-driver is the most frequently chosen mode (31.41%), followed by metro (31.16%), bus (27.13%), and car-passenger (10.30%).

San Miguel - Centro Corridor The San Miguel–Centro (SM–Centro) dataset was collected in 1985 and contains 677 individual observations. It has been extensively documented and applied in the estimation of discrete choice models (e.g., Ortúzar and Espinosa 1986; Jara-Díaz and Videla 1989;

Armstrong et al. 2001; Amador et al. 2008), and more recently, in machine learning applications (Salas et al. 2022). For the purposes of this study, we focus on the three most frequently chosen travel modes. As reported in Table 2, bus is the dominant alternative (58.25%), followed by metro (32.45%), while the remaining modes each account for less than 10% of the sample.

MNL Model Specifications

The specification of the utilities associated with each of the datasets is presented in Table 3. For the Swissmetro dataset, the utilities are identical to those used by Salas et al. (2022). Then, in Table 4, the utilities for the Las Condes - Centro dataset are specified. Also, the utilities for the San Miguel - Centro dataset are obtained by eliminating the column associated with the Car Passenger travel mode from Table 4. For more details about the explanatory variables and their definitions, we recommend that you review Salas et al. (2022).

Oversampling Methods Specification

We tuned the CVAE hyperparameters using a random search strategy over the following search space: number of units in each encoder and decoder hidden layer, batch size, number of training epochs, and the number of KL warmup epochs (the number of epochs over which the KL divergence weight is annealed from 0 to 1). To ensure robust and reproducible results, we employed a repeated stratified K-fold cross-validation procedure with 5 folds and 10 repetitions, with a fixed random seed. All random seeds were fixed simultaneously in Python, NumPy, and TensorFlow to guarantee full reproducibility.

Model selection was based on the average validation loss across cross-validation folds. Specifically, we tested 15 CVAE architectures per dataset, selected via random search, using the negative Evidence Lower Bound (ELBO) as the performance metric. The ELBO combines a reconstruction term (measuring how well the model replicates the input data) and a regularization term (penalizing divergence between the learned latent distribution and the prior via Kullback-Leibler divergence). This criterion is widely adopted for variational inference-based models.

To stabilize training, we implemented KL annealing: the weight of the KL divergence term in the ELBO is linearly increased from 0 to 1 during the first w epochs of training (the warmup period), where w is itself a hyperparameter included in the random search. This prevents the KL term from dominating the loss function in the early stages of training, allowing the encoder to learn meaningful latent representations before regularization is applied.

We found that an architecture with two hidden layers offered the best trade-off between representational capacity and convergence stability. The final selected hyperparameter values for each dataset are reported in Table 5.

For data generation and downstream travel mode choice modeling, we applied balancing exclusively within each training fold, and evaluated performance on the held-out real observations only. This procedure avoids data leakage — a critical correction relative to the original submission, where balancing was applied before partitioning, causing synthetic rows to appear in the test set and inflating the reported predictive metrics. The corrected procedure partitions on original data, augments only within the training fold, and evaluates exclusively on real observations. The reduction in reported F1 scores relative to the original submission (e.g., from 0.894 to 0.724 in SM-Centro for CVAE) reflects the magnitude of this leakage, not a deterioration in method performance.

All CVAEs were implemented with a latent space of dimension two to facilitate interpretability and controlled sample generation. We used the Adam optimizer Kingma and Ba (2014) with a learning rate of 0.001. The final selected hyperparameter values for each dataset are reported in Table 5.

The number of synthetic observations generated per dataset reflects the degree of imbalance: each minority class is oversampled up to the size of the majority class within each training fold. Table 6 reports the resulting synthetic-to-original ratios.

Note that in SM-Centro, the minority class (Car Driver, $n = 53$) requires 222 synthetic observations to reach the majority class size within each training fold, meaning that 81% of the augmented minority class consists of synthetic data. As we show in Section , this extreme ratio is directly connected to the sign reversals observed under CVAE: when the minority class is dominated by synthetic observations, any systematic distortion introduced by the generative model propagates directly to the estimated coefficients.

For ROS, SMOTE, and class-weighting we used the standard configurations implemented in Python's *Scikit-Learn* library Pedregosa et al. (2011). Class-weighting assigns each observation a weight inversely proportional to its class frequency, penalizing misclassification of minority modes more heavily during MNL estimation, without generating any synthetic data.

Quantitative Evaluation of Distributional Similarity

To quantitatively evaluate the similarity between the real and synthetic data distributions, we computed the Jensen-Shannon Divergence (JSD) for each dataset and each augmentation method. JSD values are reported per mode rather than globally, which provides a more demanding and interpretable evaluation of generative quality by isolating distributional fidelity within each class.

The results are reported in Table 7. Across all three datasets, CVAE consistently shows the largest distributional divergence. To contextualize these values, we use ROS as a practical lower-bound reference: since ROS duplicates existing observations without any transformation, it preserves the original distribution by construction and achieves near-zero JSD in SM-Centro (0.026) and Swissmetro (0.001). Against this reference, the CVAE divergence is between 8 times larger in SM-Centro (0.214 vs. 0.026) and 335 times larger in Swissmetro (0.335 vs. 0.001). In LC-Centro, all methods show higher absolute JSD values, reflecting the intrinsic complexity of that dataset's latent structure, butthe ordering is consistent across all three datasets: CVAE always departs most from the original distribution.

Dataset	Travel Mode	Frequency	Share %
Swissmetro	Train	1414	13.20
	Swiss-metro	6216	58.04
	Car	3080	28.76
SM-Centro	Car Driver (CD)	53	9.30
	Metro	185	32.45
	Bus	332	58.25
LC-Centro	Car Driver (CD)	125	31.41
	Car Passenger (CP)	41	10.30
	Metro	124	31.16
	Bus	108	27.13

Table 2. Market share for each dataset.

		Alternative		
Variables		Car	Train	SM
ASC	Constant	ASC_{Car}		ASC_{SM}
TT	Travel Time	β_{time}	β_{time}	β_{time}
Cost	Travel Cost	β_{cost}	β_{cost}	β_{cost}
Age	Age in classes		β_{age}	
GA	Annual Pass		β_{ga}	β_{ga}
Luggage	Pieces of Luggage	$\beta_{lugagge}$		
Seats	Airline seating			β_{seats}

Table 3. Swissmetro utility functions.

		Alternative			
Variables		CD	CP	Metro	Bus
ASC	Constant	ASC_{CD}	ASC_{CP}		ASC_{Bus}
TT	Travel Time	β_{time}	β_{time}	β_{time}	β_{time}
Cost	Travel Cost/wage rate	β_{cost}	β_{cost}	β_{cost}	β_{cost}
WT	Waiting Time			β_{wt}	β_{wt}
WKT	Walking Time		β_{wkt}	β_{wkt}	β_{wkt}
Sex	Sex		β_{sex}^{CP}		
AUTLIC	Cars/Driver licenses	β_{autlic}			
HH	Head of Household		β_{hh}		
MI	Monthly Income	β_{mi}^{CD}	β_{mi}^{CP}		β_{mi}^{Bus}

Table 4. Las Condes - Centro and San Migel - Centro utility functions.

Dataset	Parameter	Value
San Miguel - Centro	Hidden Units Encoder Layer 1	13
	Hidden Units Encoder Layer 2	14
	Hidden Units Decoder Layer 1	13
	Batch Size	64
	Epochs	120
	KL Warmup Epochs	10
Las Condes - Centro	Hidden Units Encoder Layer 1	13
	Hidden Units Encoder Layer 2	6
	Hidden Units Decoder Layer 1	13
	Batch Size	16
	Epochs	120
	KL Warmup Epochs	0
Swissmetro	Hidden Units Encoder Layer 1	14
	Hidden Units Encoder Layer 2	5
	Hidden Units Decoder Layer 1	9
	Batch Size	16
	Epochs	120
	KL Warmup Epochs	20

Table 5. Final hyperparameter values for the CVAE in each dataset, selected via random search with stratified 5-fold cross-validation. All architectures use ReLU activation in hidden layers and sigmoid activation in the decoder output.

These results are consistent with the sign reversal findings reported in Section . The method that most faithfully reproduces the original distributions (ROS) introduces the fewest sign reversals, while the method with the highest distributional divergence (CVAE) introduces the most. This convergence across independent analyses — JSD, predictive metrics, sign reversals, and attribute gaps — strengthens the credibility of the overall conclusion.

Dataset	Original (N)	Synthetic (N)	Ratio
SM-Centro	570	279	0.49
LC-Centro	398	102	0.26
Swissmetro	9,036	2,828	0.31

Table 6. Number of synthetic observations generated per dataset and ratio of synthetic to original observations. Balancing target: majority class size within each training fold.

Dataset	JSD ROS	JSD SMOTE	JSD CVAE
SM-Centro	0.026	0.127	0.214
LC-Centro	0.290	0.310	0.423
Swissmetro	0.001	0.008	0.335

Table 7. Mean Jensen–Shannon Divergence (JSD) between the distribution of the original data and that of the synthetic data generated by each method, computed per mode. Lower values indicate higher distributional similarity. ROS serves as a practical lower-bound reference since it preserves the original distribution by construction.

Synthetic Data Generation

The sampling for latent space elements was performed within the confidence ellipses constructed for each travel mode, as described in Section . As noted there, the ellipses serve as a practical high-density sampling mechanism, not as a solution to class separation — a motivation that would be inconsistent with the conditioning property of the CVAE. Sampling within each ellipse is performed stochastically using the multivariate normal distribution of each class, with acceptance based on Mahalanobis distance, ensuring that the density of synthetic observations follows that of the real data.

An example of the confidence ellipses generated for the SM-Centro dataset is shown in Figure 3. As can be seen, the latent representations of different classes are concentrated in overlapping regions around the origin, consistent with the expectation that the CVAE posterior does not need to separate classes. The confidence ellipses for the remaining datasets are shown in Appendix A.

Classification Comparison Results

Table 8 presents the predictive performance of the MNL model across three datasets using five methods: the unbalanced baseline (no augmentation), Conditional Variational Autoencoder (CVAE), Random Oversampling (ROS), Synthetic Minority Oversampling Technique (SMOTE), and class-weighting. Performance is evaluated using recall of the minority class, overall F1 macro, and classification accuracy. Results are based on repeated stratified 5-fold cross-validation with 10 repetitions, evaluated exclusively on real observations. Reported values correspond to the mean across all runs.

It is important to note that the predictive results reported here differ from those in the original submission due to a data leakage correction: the original procedure applied balancing before partitioning, causing synthetic observations to appear in the test set and inflating the reported metrics. The corrected procedure, described in Section , evaluates exclusively on real held-out observations. The reduction in reported F1 scores reflects the magnitude of this leakage, not a deterioration in method performance.

Additionally, we note that the holdout sample shares the same underlying correlation structure, systematic biases, and class imbalance as the training data. Consequently, good predictive performance on the holdout does not validate the model behaviorally: it confirms that the model has learned to replicate the patterns of an imbalanced dataset more effectively, but does not guarantee that the estimated parameters are economically plausible. We therefore complement the predictive analysis with a behavioral assessment in the remainder of this section.

SM-Centro. The unbalanced baseline achieves an accuracy of 0.816 but a recall of only 0.298 for the minority class (Car Driver): the model correctly identifies fewer than one in three Car Driver trips. All balancing methods improve minority class recall substantially. The CVAE achieves the highest F1 macro (0.724) and raises recall of the minority class from 0.298 to 0.601. ROS (F1: 0.718), class-weighting (F1: 0.715), and SMOTE (F1: 0.710) are competitive. This is the only dataset where CVAE is the best-performing method.

LC-Centro. The CVAE is the worst-performing method in this dataset, with an F1 macro of 0.676 — below the unbalanced baseline (0.682). Class-weighting achieves the best F1 macro (0.695) without generating any synthetic data, followed by ROS (0.694) and SMOTE (0.692). Under moderate class imbalance and greater modal diversity (four alternatives), the CVAE's generative approach does not provide a predictive advantage over simpler methods.

Swissmetro. The unbalanced baseline illustrates the most severe failure mode: it achieves an accuracy of 0.576 but a recall of 0.000 for the minority class (Train), never predicting that alternative despite it representing 13% of the sample. All balancing methods address this failure, raising F1 macro from 0.252 to approximately 0.33–0.34. SMOTE performs best (F1: 0.342), followed by ROS and class-weighting (F1: 0.335), and CVAE (F1: 0.332). The differences between methods are small, and CVAE does not outperform simpler alternatives.

Sign reversals and behavioral plausibility. Table 11 presents the estimated coefficients from the original dataset alongside the average values obtained through cross-validation for each method. A critical distinction is necessary: some sign reversals are inherited from the data and present in all methods including the baseline, while others are introduced by the augmentation method itself. Only the latter are attributable to the balancing technique.

In **SM-Centro**, the unbalanced baseline produces no sign reversals. CVAE introduces two reversals — walking time (*b_wkt*) and travel time (*b_ivt*) — not present in any other method. ROS, SMOTE, and class-weighting invert only the travel cost coefficient, which has a near-zero estimate in the original data and is therefore sensitive to small perturbations.

In **LC-Centro**, waiting time (*b_wt*) is positive in the baseline and all methods: this is a data identification issue unrelated to augmentation. However, CVAE introduces two additional reversals — in-vehicle time (*b_ivt*) and travel cost (*b_tcw*) — that are unique to that method and absent in ROS, SMOTE, and class-weighting. This is the dataset where CVAE most severely damages behavioral plausibility.

In **Swissmetro**, the travel cost coefficient is negative in the baseline and all methods. No augmentation method



Figure 3. Confidence ellipses (80%) for each alternative from the SM-Centro dataset. The modes form sub-clusters in the latent space; the Gaussian ellipse for each class spans regions of low data density, particularly for Bus, which fragments into several disconnected groups.

Dataset	Method	Accuracy	F1 Macro	Recall Minority
SM-Centro	Original (no balancing)	0.816	0.688	0.298
	CVAE	0.800	**0.724**	0.601
	ROS	0.769	0.718	0.837
	SMOTE	0.774	0.710	0.706
	Class-weighting	0.778	0.715	0.768
LC-Centro	Original (no balancing)	0.757	0.682	0.315
	CVAE	0.721	0.676	0.647
	ROS	0.740	0.694	0.584
	SMOTE	0.743	0.692	0.547
	Class-weighting	0.741	**0.695**	0.582
Swissmetro	Original (no balancing)	0.576	0.252	0.000
	CVAE	0.389	0.332	0.156
	ROS	0.390	0.335	0.169
	SMOTE	0.393	**0.342**	0.182
	Class-weighting	0.390	0.335	0.169

ROS: Random Oversampling. SMOTE: Synthetic Minority Oversampling Technique.
CVAE: Conditional Variational Autoencoder. Minority class: Car Driver (SM-Centro), Car Passenger (LC-Centro), Train (Swissmetro).

Table 8. Predictive performance of the MNL model for each dataset and method. Results based on stratified 5-fold $\times$ 10 repetitions cross-validation, evaluated exclusively on real observations. Bold indicates the best F1 macro per dataset.

introduces additional reversals: the single reversal present is a data identification issue.

Mechanism: attribute gap distortion. The sign reversals introduced by CVAE can be traced to a distortion of the structural relationship between attributes and choices in the synthetic data. For each method, we computed the mean difference in key attributes between the chosen alternative and the unchosen alternatives — the attribute gap — in the real data and in the synthetic data.

In SM-Centro, the travel time gap for the Car Driver mode inverts from $+111.8$ minutes in the real data to -37.2 minutes in the CVAE synthetic data. This means the synthetic observations show the chosen alternative as having *shorter* travel time than the unchosen ones, which is utility-inconsistent and directly reverses the identification of the travel time coefficient. Since the Car Driver mode requires 222 synthetic observations over 53 real ones (81% of the augmented minority class is synthetic), the MNL estimation is dominated by these structurally distorted observations, and the sign reversal follows directly. ROS and SMOTE preserve the attribute gaps of the original data by construction — the former by duplicating existing observations, the latter by interpolating between them — and neither introduces this distortion.

Elasticities. To make the behavioral consequences concrete, Table 9 reports the own-elasticity of travel time for the car alternative in SM-Centro under each method. In SM-Centro, the own-elasticity is negative under all methods (original: -0.180, ROS: -0.191, SMOTE: -0.202, class-weighting: -0.205), as economic theory requires. Under CVAE it becomes positive ($+0.014$): the model predicts that increasing travel time does not reduce the probability of choosing the auto, which is economically implausible. Following the approach of Li and Kockelman (2022), we also examined how predicted probabilities vary across the sample as each covariate is varied one at a time. This analysis

confirms that the positive elasticity under CVAE reflects a systematic pattern across observations, not an artifact of averaging: a substantial proportion of individuals show increasing predicted probability of choosing the car as travel time increases. This proportion is negligible under all other methods.

Method	Own-elasticity of travel time
Original (no balancing)	-0.180
CVAE	$+0.014$
ROS	-0.191
SMOTE	-0.202
Class-weighting	-0.205

Table 9. Own-elasticity of travel time for the car alternative in SM-Centro under each method. A positive value indicates that the model predicts increasing travel time raises the probability of choosing the car — an economically implausible result.

Summary. The results across the three datasets point to a conditional and asymmetric picture of CVAE performance. CVAE is competitive only in SM-Centro, where severe imbalance (9% minority share) and sufficient signal in the minority class allow the generative model to learn a useful representation. In LC-Centro, CVAE underperforms the unbalanced baseline in predictive terms and introduces the largest number of sign reversals. In Swissmetro, CVAE provides modest predictive improvements similar to other methods but does not outperform SMOTE. Across all three datasets, the behavioral damage introduced by CVAE is concentrated in the datasets and classes where the synthetic-to-real ratio is highest — confirming that the distortion mechanism identified in the attribute gap analysis is the primary driver of sign reversals.

Sensitivity analysis: confidence ellipse level. To assess whether the choice of 80% confidence ellipse is driving the results, we repeated the generation and evaluation procedure with ellipse levels of 70%, 80%, 90%, and 95%. Table 10 reports the F1 macro of the CVAE under each level. The results are robust: F1 macro varies by at most 0.007 in SM-Centro and by less than 0.001 in LC-Centro and Swissmetro. A consistent pattern is that the 70% level yields a slightly higher F1, coherent with the interpretation that a smaller ellipse restricts sampling to the densest region of the latent space, avoiding peripheral areas where the CVAE generates lower-quality observations. Overall, the choice of 80% does not drive the conclusions of the paper.

	Confidence Level			
Dataset	**0.70**	**0.80**	**0.90**	**0.95**
SM-Centro	0.731	0.724	0.725	0.727
LC-Centro	0.676	0.676	0.675	0.676
Swissmetro	0.331	0.332	0.332	0.332

Table 10. F1 macro of the CVAE under different confidence ellipse levels. Results are robust to this design choice across all three datasets.

Discussions and Conclusions

Class imbalance in travel mode choice data is a persistent challenge in transportation behavior modeling. When some travel modes are much less frequently observed than others, models like the MNL tend to produce biased estimates and poor predictive performance for minority classes, dominated by the majority. Addressing this challenge requires methods that not only improve minority class prediction but also preserve the behavioral plausibility of the estimated parameters. As this study shows, generative data augmentation does not always meet this requirement.

The main objective of this study was to evaluate the use of the Conditional Variational Autoencoder (CVAE) as a data augmentation strategy for travel mode choice modeling under class imbalance, and to compare it against ROS, SMOTE, and class-weighting. Our results reveal a conditional and asymmetric picture that diverges substantially from the original submission's conclusions.

On predictive performance, CVAE improves F1 macro over the unbalanced baseline only in two of three datasets, and is the best method in only one (SM-Centro, F1: 0.724). In LC-Centro, CVAE is the worst-performing method, falling below the unbalanced baseline (F1: 0.676 vs. 0.682). In Swissmetro, all balancing methods provide similar modest improvements, with SMOTE performing best (F1: 0.342) and CVAE slightly behind (F1: 0.332). Class-weighting, which requires no synthetic data generation, achieves the best predictive performance in LC-Centro and is competitive in the other two datasets, suggesting that simpler methods should be considered before adopting generative approaches.

On behavioral plausibility, the most significant finding of this study is the occurrence of sign reversals in key behavioral parameters under CVAE in two of three datasets. In SM-Centro, CVAE introduces reversals in travel time and walking time coefficients not present in any other method. In LC-Centro, it introduces additional reversals in in-vehicle time and travel cost. These reversals are not inherited from the original data but are specific to the CVAE augmentation method, and they render the resulting model economically implausible for policy analysis. We trace these reversals to a concrete mechanism: the distortion of attribute gaps between chosen and unchosen alternatives in the synthetic data. In SM-Centro, the travel time gap for the Car Driver mode inverts from $+111.8$ minutes in the real data to -37.2 minutes in the CVAE synthetic data, directly propagating to the sign reversal in the estimated coefficient. This distortion is absent under ROS, SMOTE, and class-weighting, which either preserve the original observations or interpolate between them.

The elasticity analysis makes the behavioral consequences concrete. In SM-Centro, the own-elasticity of travel time for the auto alternative is negative under all methods except CVAE, where it becomes positive ($+0.014$). Following Li and Kockelman (2022), a sample-level analysis confirms that this reflects a systematic pattern across observations: under CVAE, a substantial proportion of individuals show increasing predicted probability of choosing the car as travel time increases, an economically implausible result not observed under any other method.

		ORIG.		ROS		SMOTE		CVAE		Class-w.	
Dataset		$\widehat{\beta}$	S.E.	$\overline{\widehat{\beta}}$	S.D.$_{CV}$	$\overline{\widehat{\beta}}$	S.D.$_{CV}$	$\overline{\widehat{\beta}}$	S.D.$_{CV}$	$\overline{\widehat{\beta}}$	S.D.$_{CV}$
SM-Centro	*Car Driver ASC*	-5.169	1.052	-4.153	0.451	-4.610	0.474	-6.177	0.813	-4.821	0.412
	Bus ASC	-1.896	0.586	-1.743	0.263	-2.240	0.313	-1.122	0.429	-1.987	0.271
	Travel Time	-0.001	0.001	-0.002	0.000	-0.001	0.000	+0.004	0.000	-0.002	0.000
	Walking Time	-0.001	0.000	-0.001	0.000	-0.001	0.000	+0.002	0.000	-0.001	0.000
	Waiting Time	-0.006	0.003	-0.002	0.001	-0.005	0.001	+0.001	0.001	-0.003	0.001
	Travel Cost	-0.001	0.012	+0.013	0.005	+0.017	0.005	+0.040	0.009	+0.011	0.004
	AUTLIC	+1.248	0.720	+1.653	0.295	+1.565	0.282	+2.619	0.498	+1.589	0.274
LC-Centro	*Car Driver ASC*	-4.268	0.752	-4.958	0.336	-4.961	0.404	-3.728	0.339	-4.743	0.318
	Car Pass. ASC	-3.167	0.613	-2.495	0.218	-2.737	0.323	+0.182	0.270	-2.613	0.201
	Bus ASC	-0.985	0.588	-0.754	0.348	-1.147	0.403	-1.854	0.439	-0.891	0.332
	Travel Time	-0.002	0.001	-0.002	0.000	-0.002	0.000	+0.003	0.000	-0.002	0.000
	Walking Time	-0.002	0.001	-0.003	0.000	-0.003	0.000	-0.001	0.000	-0.003	0.000
	Waiting Time	+0.003	0.003	+0.003	0.002	+0.002	0.002	+0.013	0.002	+0.002	0.002
	Travel Cost	-0.003	0.012	-0.001	0.007	-0.002	0.007	+0.036	0.006	-0.002	0.007
	SEX (CP)	-0.881	0.434	-0.977	0.125	-1.660	0.147	-0.993	0.157	-1.021	0.119
	AUTLIC	+2.450	0.641	+2.981	0.293	+3.028	0.315	+3.328	0.339	+2.897	0.281
Swissmetro	*Car ASC*	+1.370	0.094	+0.661	0.025	+0.376	0.033	+0.712	0.032	+0.598	0.024
	SM ASC	+1.537	0.093	+0.121	0.025	-0.354	0.030	+0.301	0.036	+0.089	0.024
	Travel Cost	-0.001	0.000	-0.001	0.000	-0.001	0.000	-0.001	0.000	-0.001	0.000
	Travel Time	-0.005	0.000	-0.004	0.000	-0.005	0.000	-0.007	0.000	-0.004	0.000
	Age	+0.049	0.028	+0.020	0.007	-0.093	0.010	+0.155	0.010	+0.031	0.006
	Luggage	-0.336	0.039	-0.444	0.018	-0.684	0.020	-0.061	0.017	-0.421	0.017
	GA	+7.664	0.276	+8.169	0.137	+8.271	0.136	+9.006	0.148	+8.054	0.131
	Seats	+0.300	0.066	-0.049	0.032	+0.747	0.030	+0.197	0.028	-0.031	0.030

ORIG.: original dataset (S.E. = standard error of the MLE).

S.D.$_{CV}$: standard deviation of $\overline{\widehat{\beta}}$ across cross-validation repetitions.

These two quantities are conceptually distinct: S.E. reflects estimation uncertainty for a fixed dataset,

while S.D.$_{CV}$ reflects instability across random data splits. Coefficients whose sign differs from ORIG. are shown in bold in the text discussion.

ROS: Random Oversampling. SMOTE: Synthetic Minority Oversampling. CVAE: Conditional Variational Autoencoder.

Class-w.: class-weighting (inverse frequency weights).

Table 11. Stability and variability of estimated parameters across sampling methods. Sign reversals relative to the original estimates are highlighted in the text.

These findings highlight a fundamental trade-off. The benefit of CVAE depends on a combination of sufficient class imbalance and sufficient signal in the minority class for the generative model to learn a useful representation. When the minority class requires a very high ratio of synthetic to real observations (81% in SM-Centro for Car Driver), any systematic distortion introduced by the generative model dominates the augmented dataset and propagates to the estimated coefficients. The distributional analysis confirms this: CVAE consistently shows the largest Jensen-Shannon Divergence from the original data across all datasets, up to 335 times larger than ROS in Swissmetro. The method that most faithfully reproduces the original distributions (ROS) also introduces the fewest sign reversals; the method with the highest distributional divergence (CVAE) introduces the most.

Practical recommendations. Based on these findings, we offer the following guidance for practitioners:

- CVAE should be considered only when the primary objective is predictive performance and class imbalance is severe. Under moderate imbalance, simpler methods such as class-weighting or ROS provide competitive predictive performance without the behavioral risks.
- Before using CVAE-augmented data for any policy application, the coherence of the estimated coefficients must be verified explicitly. Sign reversals in travel time, cost, or waiting time coefficients indicate that the augmented data has distorted the structural attribute–choice relationship, and the resulting model should not be used for behavioral inference or policy analysis.
- The synthetic-to-real ratio is a critical parameter. When a minority class requires a very high proportion of synthetic observations to reach balance, the risk of behavioral distortion increases substantially. Practitioners should report this ratio and consider whether the degree of augmentation is appropriate for their application.

Limitations and future research. One limitation of this study is the restricted exploration of how the oversampling ratio influences model performance. While we balanced each minority class to the size of the majority class, varying this ratio could affect both predictive accuracy and parameter stability. Future studies should systematically investigate optimal oversampling levels for different datasets and modeling goals.

The CVAE architectures explored here are relatively simple (two hidden layers, small latent space). Future research could explore more advanced architectures — such as hierarchical CVAEs or those incorporating attention mechanisms — that may generate higher-quality synthetic data with fewer structural distortions. The integration of CVAE with Generative Adversarial Networks (GANs) is another promising direction. Additionally, the use of

multimodal data sources within CVAE-based frameworks could improve minority class representation by leveraging richer behavioral and contextual information.

Finally, while random search provided adequate hyperparameter selection for the architectures considered here, Bayesian optimization methods such as those implemented in Optuna could yield better configurations with fewer evaluations, particularly for larger datasets or more complex architectures.

Implications for Transportation Policy

Class imbalance in mobility data has direct implications for transportation planning and policy-making. When models systematically underestimate the demand for less common travel modes, this may lead to underinvestment in infrastructure, misallocation of resources, and the neglect of specific population needs. The findings of this study suggest that addressing class imbalance through data augmentation can improve minority mode prediction, but must be accompanied by careful verification of behavioral plausibility — particularly when the resulting models are intended to inform investment decisions or policy design. Methods that preserve the structural attribute–choice relationship of the original data, such as ROS and class-weighting, offer a more reliable foundation for policy analysis than generative methods that may distort this relationship.

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Author Contributions

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Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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